The US-China Trade War and Global Reallocations
Fajgelbaum, Goldberg, Kennedy, Khandelwal, and Taglioni
American Economic Review: Insights, 2023

Replication Instructions

This folder contains data files and codes to replicate the empirical results in the paper. Authors obtained the results using Stata 18. Users may need to install the Stata packages listed below. The raw data files used in the analysis are large, and implementation may require significant computing resources and memory. Run-time of the full replication from start to finish is approximately 2-3 hours.

Stata packages: gtools; carryforward; wbopendata; ftools; estout; reghdfe; binscatter; coefplot; net install dm0082 from("http://www.stata-journal.com/software/sj15-3"); net install grc1leg, from("http://www.stata.com/users/vwiggins/")

Computing environment of the replicator:

- CCSS Shared Windows Server 2022 Datacenter, 256GB, Intel Zeon E5-4669 v3 @ 2.10Ghz (4 processors, 36 cores)
- Stata/MP 18

For bootstrap code:

- BioHPC Linux server, Rocky Linux 9.0, AMD Opteron Processor 6380 / 64 cores/ 256 GB memory
- Stata/MP 16

To run the full replication code:

1. Download UN Comtrade data for years 2014-2019, one file per year, and save the files in .dta format with the corresponding year name in the raw/data/comtrade/ directory (for example, "2014.dta"). Users can follow the bulk download or API instructions on the Comtrade website and should download the following variables for all countries and where commodity codes are 6-digit HS codes: year, reportercode, partnercode, commoditycode, imports_q, imports_nw, imports_v, exports_q, exports_nw, exports_v, reporteriso, partneriso.
2. Set the root directory in line 9 of **code/00_directories.do**, and run this program. All other directory paths will be set automatically.
3. Copy the root directory above into line 19 of **code/bsregs.do** (but do not run the program)
4. Run **code/01_master.do**, which implements all the cleaning and analysis programs necessary to reproduce the empirical results. The **01_master.do** file and its subsidiary programs include additional documentation describing the procedures implemented and the output generated from each do file.

To run only the analysis programs (but not the data cleaning programs), complete steps 1-2 above and then run **code/01_master.do** starting from line 48.

Note that line 52 of **code/01_master.do** calls a shell program, which implements the bootstrap procedure using parallel processing. If parallel processing is not available, the following alternate procedure is also sufficient:

- Comment out line 8 of **code/bsregs.do**
- Un-comment the loop in lines 10 and 342 of **code/bsregs.do**
- Run **code/bsregs.do** in lieu of the shell program

However, running the loop in lieu of parallel processing significantly increases total run-time.

Statement About Rights

- We certify that the authors of this manuscript have legitimate access to and permission to use the data used in this manuscript.
- We certify that the authors of this manuscript have documented permission to redistribute/publish the data contained within this replication package.

Descriptions of Raw Data Sets

Paper bibliography includes detailed citations for all raw data sources.

- **/Bown_etal/** --- data from Bown et al. (2019) of tariff lines on which China decreased MFN tariffs for countries other than the US, as well as product lines on which car parts were exempt from tariffs. Data are publicly available here as of November 16, 2023.
- **/census_trade/** -- includes data from Census and Fajgelbaum et al. (2020) on import and export flows and tariffs at the country-product-month level, where tariff lines were extended to December 2019 by the authors. Merge of data from Census and data from Fajgelbaum et al. (2020) was performed manually by the authors. For information on how to access the publicly available Census trade data, see here (accessed as of November 16, 2023).
- **/AKW/** -- Chinese trade flow data from Ahn, Khandelwal, and Wei (2011). Data are available from amit.khandelwal@yale.edu.
- **/comtrade/** -- These data cannot be shared by the authors due to restrictions on the terms of use by UN Comtrade. Data however can be downloaded from the UN comtrade website here. Authors accessed the data on or before November 1, 2022.
- **/crosswalks/** -- classification of HS trade codes into broad sectors and over-time harmonization of HS codes, also from the UN
- **/deep_trade/** -- data on deep trade agreements from Mattoo et al. (2020). Data publicly available here as of November 16, 2023.

- **/distances/** -- data on bilateral country distances from Thierry Mayer and Soledad Zignago and publicly available via CEPII (see citation in the bibliography). Data are available under a Etalab 2.0 license here (see the "GeoDist" database). Authors accessed the data on or before November 1, 2022.
- **/retaliatory_tariffs/** -- data on retaliatory tariff increases by US trade partners, from Fajgelbaum et al. (2020) and extended to December 2019 by the authors. Data are publicly available per the authors.
- **/unctad/** -- data on country GDP and FDI flows and stocks, from UNCTAD, publicly available here as of November 16, 2023.
- **/wb_country_groups/** -- country codes from the World Bank. Data are publicly available and can be downloaded here as of November 16, 2023.

Data Processing Programs

Downstream data files are output into the **data/processed/** folder.

Trade flows

- **comtrade.do** – Cleans the comtrade data; produces files called global_trade*.dta

Country codes and export rankings

- **iso_codes.do** – Creates dataset of ISO country codes, iso_codes.dta
- **cty_codes.do** -- Creates dataset of Census country codes, cty_codes2.dta
- **cty_rank.do** – Creates dataset of countries ranked by their global export value from 2015-17, cty_rank2.dta

Tariffs

- **china_weight.do** – Creates dataset of weights used to collapse tariffs to the HS6-level using pre-trade war flow values, china_weight_hs6.dta
- **z_usch.do** – Creates dataset of US tariffs on China, z_usch.dta
- **z_chus.do** – Creates dataset of Chinese tariffs on the US, z_chus.dta
- **z_chrw.do** – Creates dataset of Chinese MFN tariff cuts on RW, z_chrw.dta
- **z_usi.do** – Creates dataset of US tariffs on non-China countries, z_usi.dta

Country characteristics

- **cty_distances.do** – Creates dataset of bilateral country distances, cty_distances_US_CN.dta
- **FDI_stock.do** – Creates dataset of country FDI stocks 2016-18, cty_FDI_stock.dta
- **FBI_inflow.do** – Creates dataset of country FDI inflows 2016-18, cty_FDI_flow.dta
- **country_gdp.do** – Creates dataset of countries' GDP merged with other characteristics, country_gdp_dist.dta
- **chars.do** – Merges data on country characteristics, creates chars.dta

Data Build

- **build.do** – Merges all datasets and creates the two datasets used in analysis: data5.dta and rf.dta

<u>Data Analysis Programs</u>

All figures and tables are output into the /**results/** folder.

- **didplots.do** – Creates Figures 1, A3, and A4, and Table A2
- **shbsregs.sh** – Shell program runs main regressions + bootstraps in parallel processing
- **bsregs.do** – An alternative program for running the bootstraps if parallel processing is not available (takes longer)
- **winners.do** – Creates Figures 2 and 3
- **quadplot.do** – Creates Figure 4
- **sumplots.do** – Creates Figures A1 and A2

Data Preparation Program	Dataset created
comtrade.do	global_trade*.dta
iso_codes.do	iso_codes.dta
cty_codes.do	cty_codes2.dta
cty_rank.do	cty_rank2.dta
china_weight.do	china_weight_hs6.dta
z_usch.do	z_usch.dta
z_chus.do	z_chus.dta
z_chrw.do	z_chrw.dta
z_usi.do	z_usi.dta
cty_distances.do	cty_distances_US_CN.dta
FDI_stock.do	FDI_stock.dta
FBI_inflow.do	cty_FDI_flow.dta
country_gdp.do	country_gdp_dist.dta
chars.do	chars.dta
build.do	data5.dta rf.dta

Figure/Table #	Program	Line Number
Figure 1	didplots.do	317
Figure 2	winners.do	86
Figure 3	winners.do	149
Figure 4	quadplot.do	187
Figure A.1	sumplots.do	67
Figure A.2	sumplots.do	92, 101
Figure A.3	didplots.do	321
Figure A.4	didplots.do	325
Table A.1	not empirical	-
Table A.2	didplots.do	336

References

Bown, C., E. Jung, and E. Zhang (2019a). Data for: Trump Has Gotten China to Lower Its Tariffs. Just Toward Everyone Else. https://www.piie.com/blogs/trade-and-investment-policy-watch/ trump-has-gotten-china-lower-its-tariffs-just-toward.

Fajgelbaum, P. D., P. K. Goldberg, P. J. Kennedy, and A. K. Khandelwal (2020a). Data for: The return to protectionism. https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10. 7910/DVN/KSOVSE.

Mattoo, A., N. Rocha, and M. Ruta (2020a). Data for: Handbook of deep trade agreements. https://datatopics.worldbank.org/dta/table.html

Mayer, T. and S. Zignago (2011, December). Notes on CEPII distances measures: The GeoDist database. Working Papers 2011-25, CEPII Research Center.

UN Comtrade (2022). UN comtrade database. http://comtrade.un.org/

UNCTAD (2022). Investment statistics and trends. https://unctad.org/topic/investment/investment-statistics-and-trends.